

Exercises Set: Combinatorics

Solutions

DSBA Mathematics Refresher 2026

1 Company employees (*)

Write M for the set of men, U for the set of union members and W for the set of married employees ("wedded", to avoid a second M). The data is

$$|M| = 300, \quad |U| = 352, \quad |W| = 424$$

$$|U \cap M| = 188, \quad |W \cap M| = 166, \quad |U \cap W| = 208, \quad |U \cap W \cap M| = 144$$

and the total number of employees is 800.

A *single, non-union woman* is an employee who is in **none** of the three sets, so the answer is

$$800 - |M \cup U \cup W|$$

Method 1: inclusion-exclusion. The formula for three sets is

$$|M \cup U \cup W| = |M| + |U| + |W| - |U \cap M| - |W \cap M| - |U \cap W| + |U \cap W \cap M|$$

Why this formula? Adding the three sets counts the employees belonging to exactly two of them twice, so we subtract the three pairwise intersections; but the employees belonging to all three, counted 3 times and then removed 3 times, have disappeared altogether, so we add them back once.

Numerically:

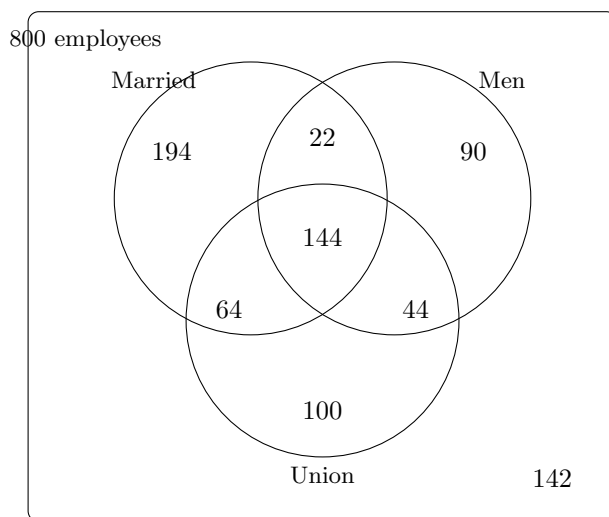
$$|M \cup U \cup W| = 300 + 352 + 424 - 188 - 166 - 208 + 144 = 1076 - 562 + 144 = 658$$

so the answer is

$$800 - 658 = \boxed{142}$$

Method 2: filling in a Venn diagram. It is instructive (and safer) to compute every region, starting from the middle and working outwards. Each time, we subtract from a given group the parts of it that we have already placed.

- All three: 144.
- Union men, *not* married: $188 - 144 = 44$.
- Married men, *not* in the union: $166 - 144 = 22$.
- Married union members, *not* men: $208 - 144 = 64$.
- Men only: $300 - (144 + 44 + 22) = 90$.
- Union only: $352 - (144 + 44 + 64) = 100$.
- Married only: $424 - (144 + 22 + 64) = 194$.



The seven regions add up to

$$144 + 44 + 22 + 64 + 90 + 100 + 194 = 658$$

and therefore

$$800 - 658 = 142$$

There are 142 single women who are not in the union.

2 The padlock

The code is an ordered sequence of 4 digits, each from 0 to 9, and digits may be repeated unless stated otherwise. Every question below is solved with the **multiplication principle**: if a choice is made in successive independent steps, the number of possibilities is the product of the number of possibilities at each step.

1. How many possible codes? There are 10 possibilities for each of the 4 digits:

$$10 \times 10 \times 10 \times 10 = 10^4 = \boxed{10\,000}$$

2. Codes with 4 different digits. Now each digit used is no longer available for the following positions:

1st digit: 10 possibilities
 2nd digit: 9 possibilities (all but the first)
 3rd digit: 8 possibilities
 4th digit: 7 possibilities

$$10 \times 9 \times 8 \times 7 = \frac{10!}{6!} = \boxed{5\,040}$$

This is the number of **arrangements** (ordered selections without repetition) of 4 digits among 10.

3. Codes ending with an even number. The last digit must be one of 0, 2, 4, 6, 8, so 5 possibilities, and the first three are free:

$$10 \times 10 \times 10 \times 5 = \boxed{5\,000}$$

which is indeed half of the total: exactly one code out of two ends with an even digit.

4. Codes ending with an even number, with 4 different digits. Here the order of the choices matters for the *counting* (not for the code): we start with the constrained position.

4th digit: 5 possibilities (it must be even)
 1st digit: 9 possibilities (any digit but the last one)
 2nd digit: 8 possibilities
 3rd digit: 7 possibilities

$$5 \times 9 \times 8 \times 7 = \boxed{2\,520}$$

Remark: had we started with the first digit, we would have been stuck — the number of possibilities left for the last digit would depend on whether the digits already used were even or odd. **Always deal with the constrained positions first.**

5. Codes containing at least one digit 4. Counting these directly is painful (there may be one, two, three or four 4's, in any positions). It is much easier to count the **opposite**: the codes containing *no* 4 at all. For those, each of the 4 positions has 9 possibilities (every digit but 4), so there are $9^4 = 6\,561$ of them, and

$$\underbrace{10^4}_{\text{all codes}} - \underbrace{9^4}_{\text{without any 4}} = 10\,000 - 6\,561 = \boxed{3\,439}$$

"At least one" is almost always a signal to count the complement.

6. Codes containing exactly one digit 4. We choose where the 4 goes, then fill the rest:

- 4 possibilities for the position of the digit 4;
- 9 possibilities for each of the 3 remaining positions (any digit but 4, since we want *exactly* one).

$$4 \times 9^3 = 4 \times 729 = \boxed{2\,916}$$

Check: $2\,916 < 3\,439$, as it must be — "exactly one" is a particular case of "at least one". The difference, 523, counts the codes with two, three or four 4's.

3 Airplanes Company

The 4 airplanes are distinguishable, so we assign a crew to each of them in turn.

The pilots. Assigning 4 distinct pilots to 4 distinct airplanes is choosing an order for the pilots:

$$4! = 24 \text{ ways}$$

The flight attendants. We must split 8 flight attendants into 4 pairs, one per airplane. Here, the order *within* a pair does not matter (the two attendants of a plane are simply "the crew of that plane"), so we use **binomial coefficients**:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

which counts the number of ways of choosing k objects among n *without order*.

1st airplane: $\binom{8}{2} = 28$ ways (choose 2 attendants among the 8 available)
 2nd airplane: $\binom{6}{2} = 15$ ways
 3rd airplane: $\binom{4}{2} = 6$ ways
 4th airplane: $\binom{2}{2} = 1$ way (the two who are left)

$$\binom{8}{2} \binom{6}{2} \binom{4}{2} \binom{2}{2} = 28 \times 15 \times 6 \times 1 = 2\,520 \text{ ways}$$

Total. The two choices are independent, so

$$4! \times \binom{8}{2} \binom{6}{2} \binom{4}{2} \binom{2}{2} = 24 \times 2\,520 = \boxed{60\,480}$$

Remark: why the binomial coefficient looks like that. To choose k objects among n *in order*, there are $n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$ possibilities. But each unordered selection of k objects has been counted once for each of its $k!$ orderings, so we divide by $k!$, which gives $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

This also gives a quicker route to the 2 520 above: line up the 8 attendants ($8!$ ways) and give the first two to plane 1, the next two to plane 2, and so on. Each assignment is obtained $2!$ times per airplane (the two members of a crew can be swapped), hence

$$\frac{8!}{2!2!2!2!} = \frac{40\,320}{16} = 2\,520 \checkmark$$

4 Bookshelf (*)

There are 4 mathematics books, 6 physics books and 3 chemistry books, all distinct: 13 books in total.

If the books are grouped by subject. Two independent things have to be decided:

- the order of the three *groups* on the shelf: $3! = 6$ ways;
- the order of the books *inside* each group: $4!$ for mathematics, $6!$ for physics, $3!$ for chemistry.

$$3! \times 4! \times 6! \times 3! = 6 \times 24 \times 720 \times 6 = \boxed{622\,080}$$

If only the mathematics books are grouped. The trick is to treat the mathematics block as a **single object**. We then have to order

$$\underbrace{1}_{\text{the maths block}} + \underbrace{6}_{\text{physics}} + \underbrace{3}_{\text{chemistry}} = 10 \text{ objects}$$

which can be done in $10!$ ways; and inside the block, the 4 mathematics books can be ordered in $4!$ ways:

$$4! \times 10! = 24 \times 3\,628\,800 = \boxed{87\,091\,200}$$

Check: this is much larger than the previous answer, which is reassuring — the second situation is far less constrained than the first.

5 ESSEC's Stairs (*)

Let $F(n)$ be the number of ways of climbing a staircase of n steps, going up either 1 or 2 steps at a time. Let us compute the first values by hand:

n	$F(n)$	the different climbs
0	1	do nothing
1	1	1
2	2	1 + 1 ; 2
3	3	1 + 1 + 1 ; 2 + 1 ; 1 + 2
4	5	1 + 1 + 1 + 1 ; 2 + 1 + 1 ; 1 + 2 + 1 ; 1 + 1 + 2 ; 2 + 2
5	8	...
6	13	...

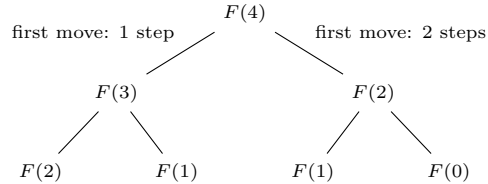
These are the Fibonacci numbers — but *why*?

The recurrence. Consider any way of climbing n steps, and look only at **the first move**. There are two possibilities, and they exclude each other:

- the first move is a single step; there remain $n - 1$ steps to climb, in $F(n - 1)$ ways;
- the first move is a double step; there remain $n - 2$ steps to climb, in $F(n - 2)$ ways.

Every climb falls into exactly one of these two cases, so we may add the two counts:

$$F(n) = F(n - 1) + F(n - 2), \quad \text{with } F(0) = 1 \text{ and } F(1) = 1$$



Each climb is sorted into exactly one of the two branches according to its first move, and the same question then reappears on a shorter staircase — which is what makes the count recursive.

This is precisely the Fibonacci sequence.

The answer. We now only have to iterate up to $n = 16$:

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
$F(n)$	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	987	1597

$$F(16) = \boxed{1\,597}$$

Remark: note how the "first move" argument replaced a hopeless enumeration by two smaller versions of the same problem. Splitting a count into exclusive cases (here, by conditioning on the first move) is one of the most useful techniques in combinatorics.